ENERGY CONSUMPTION ESTIMATION SYSTEM USING MACHINE LEARNING

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Abstract

Energy consumption forecasting plays an important role in efficient energy management and smart home automation systems. Accurate prediction of electricity usage helps reduce energy wastage and improves the overall efficiency of electrical systems.

This project presents a machine learning-based system that predicts household appliance electricity consumption using environmental parameters such as temperature and humidity. The system uses a **Random Forest Regression model** trained on historical energy consumption data.

A web interface built using **Flask** allows users to train the machine learning model, upload datasets, and predict energy consumption based on environmental conditions. The developed system demonstrates how machine learning techniques can be used to build intelligent solutions for energy management and smart homes.

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1. Introduction

Energy consumption prediction is an important area of research in modern smart grid systems and intelligent home automation. With the increasing number of electrical devices used in households, predicting electricity demand has become essential for optimizing energy usage and reducing wastage.

Machine learning techniques allow computers to learn patterns from historical data and use that knowledge to predict future outcomes. By analyzing environmental conditions such as temperature and humidity, it is possible to estimate how much electricity appliances will consume under certain conditions.

This project focuses on building a machine learning system capable of predicting household appliance energy consumption. The system includes a user-friendly web interface that allows users to train the model and generate predictions easily.

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2. Problem Statement

The objective of this project is to develop a machine learning system that can **predict household appliance electricity consumption** based on environmental conditions such as temperature and humidity.

The system should:

- Train a machine learning model using historical energy consumption data
- Allow users to upload custom datasets
- Predict energy usage based on environmental input parameters
- Display prediction results through a web-based interface

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3. Objectives of the Project

The main objectives of the project are:

- To analyze environmental data related to household energy consumption
- To develop a machine learning model capable of predicting appliance electricity usage
- To evaluate model performance using evaluation metrics
- To design a web interface that allows users to interact with the prediction system
- To support dynamic dataset upload and real-time predictions

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4. Dataset Source and Description

Dataset Name

Appliances Energy Prediction Dataset

Dataset Source

The dataset used in this project is obtained from the **UCI Machine Learning Repository**, which is a widely used public database for machine learning research datasets.

Dataset Link:
https://archive.ics.uci.edu/ml/datasets/Appliances+energy+prediction

Dataset Description

The dataset contains measurements of **energy consumption of household appliances along with environmental conditions such as temperature and humidity inside a smart home environment**.

The data was collected using **IoT sensors installed in different rooms of a residential house**. These sensors continuously recorded temperature, humidity, and electricity consumption values.

This dataset is commonly used for research in:

- Smart home energy management

- Energy demand prediction
- Machine learning-based energy forecasting

Features Used in the Project

Feature	Description
T1	Temperature in Room 1 (°C)
RH_1	Humidity in Room 1 (%)
T2	Temperature in Room 2 (°C)
RH_2	Humidity in Room 2 (%)
Appliances	Energy consumption of appliances (Wh)

Target Variable

The **Appliances** column represents the electricity consumption of household appliances measured in **Watt-hours (Wh)**.

The machine learning model predicts this value using environmental features.

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5. Methodology

The system follows a standard machine learning workflow consisting of several stages.

1. Data Collection

The dataset is obtained from the UCI Machine Learning Repository.

2. Data Preprocessing

Relevant columns are selected and unnecessary features are removed.

3. Feature Selection

The following environmental variables are used as input features:

- Temperature (T1, T2)
- Humidity (RH_1, RH_2)

4. Data Splitting

The dataset is divided into:

- **80% Training Data**

- **20% Testing Data**

5. Model Training

A **Random Forest Regression model** is trained using the training dataset.

6. Model Evaluation

The model performance is evaluated using the **R^2 Score**, which measures how well the predicted values match the actual energy consumption.

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6. Machine Learning Model

The project uses the **Random Forest Regression algorithm**.

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

Advantages of Random Forest

- Handles nonlinear relationships effectively
- Reduces model overfitting
- Works well with large datasets
- Produces accurate predictions

The model learns the relationship between environmental conditions and energy consumption to generate predictions.

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7. System Architecture

The system consists of three main components.

Frontend

The frontend is developed using **HTML and CSS** and provides a simple interface for users to interact with the system.

Backend

The backend is developed using **Flask (Python)**. It handles dataset processing, model training, and prediction generation.

Machine Learning Model

The trained Random Forest model processes user input and generates energy consumption predictions.

System Flow:

User Input → Flask Backend → Machine Learning Model → Prediction Output

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8. Implementation

The project is implemented using the following technologies:

Technology	Purpose
Python	Programming language
Flask	Backend web framework
Pandas	Data processing
Scikit-learn	Machine learning model
HTML/CSS	Frontend interface

The system allows users to:

1. Train the model using a predefined dataset
2. Upload custom datasets
3. Enter environmental parameters
4. Predict appliance electricity consumption

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9. Results and Evaluation

The model performance is evaluated using the **R^2 score**, which measures how well the predicted values match the actual energy consumption values.

An R^2 score closer to **1** indicates better prediction accuracy.

The trained Random Forest model successfully predicts household appliance electricity consumption based on temperature and humidity conditions.

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10. Conclusion

This project demonstrates the application of machine learning techniques for predicting energy consumption in residential environments. By analyzing environmental factors such as temperature and humidity, the system can estimate electricity usage of household appliances.

The developed web-based interface allows users to train the model and generate predictions easily. This system can contribute to improving energy efficiency and supporting smart home energy management systems.

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11. Future Scope

The system can be further improved by:

- Using larger real-world datasets
- Adding additional environmental features such as weather conditions
- Implementing advanced machine learning models
- Deploying the system as an online application
- Integrating with IoT smart home devices

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12. References

1. UCI Machine Learning Repository – Appliances Energy Prediction Dataset
2. Scikit-learn Documentation
3. Flask Web Framework Documentation
4. Research papers on energy consumption prediction